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Engineering Portfolio

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I am a student at Northeastern University pursuing a degree in Mechanical Engineering with a minor in Computer Science. My interests include designing innovative mechanical systems, optimizing mechanical processes, utilizing CAD software, using 3D printing to develop prototypes, and solving complex engineering challenges.

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Mechanical Engineering co-op on a team focused on laboratory automation for lipid nanoparticle research

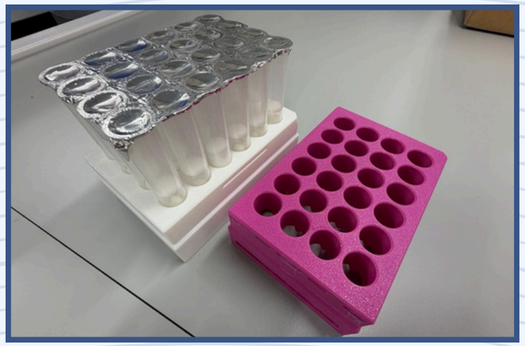
Background

- First co-op (July-Dec 2025) at Moderna, an mRNA vaccine company
- Lipid Nanoparticles (LNPs) protect mRNA and deliver it to cells
- My co-op team is developing a work cell for end-to-end automation of these experiments, improving throughput and consistency

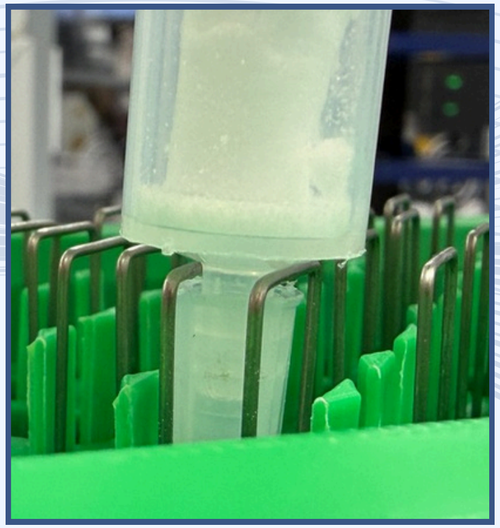
Overview

- Primary responsibility was the development of a fully automated system for preparing size exclusion columns (SEC) for robotic handling
  - Important consumable in LNP process
- Device had to automate storage, decapping, transport, and cleanup
- Integration with larger system required
- Rapid testing and iteration to improve reliability was key to meet automation goals
- Sourced components and used 3D Printing

Initial fixture design for removing caps from racks of SEC columns. A robot arm interacts with the device from the back and controls the pneumatic actuators which remove the caps.



Custom rack for 24 columns to fit standardized labware



Close-up of spring-based cap removal mechanism

Applied Technologies Co-op | Moderna

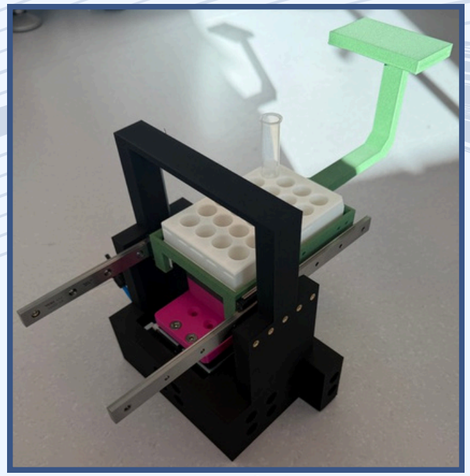
Custom heavy duty drawer for storing equipment under a lab bench.

Core Skills and Takeaways

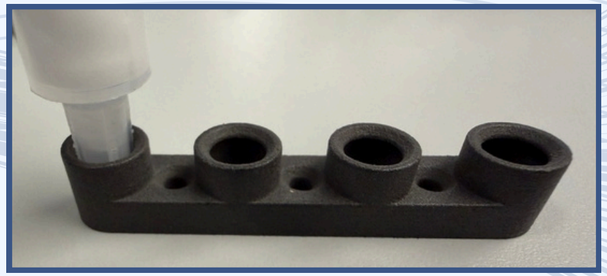
Rapid prototyping and iteration
SOLIDWORKS and 3D printing
Sourcing components
Creating a Bill of Materials
Testing for reliability
Integrating robotic systems
Working with crossfunctional engineering teams

More Details

- Other responsibilities included general engineering support for team members
  - Designed and built heavy duty custom under-bench drawer for lab equipment
  - Created Bill of Materials and prioritized standardized components
- Worked on version 2 of cap remover at end of co-op to improve reliability and preparation speed
  - Used barb-based mechanism instead of springs
  - Incorporated lessons from version 1



A prototype of the Version 2 cap remover



Close-up of barb mechanism

Northeastern University Robotics Club

Projects include a custom robot arm with a hand-style end effector and a 3D printer designed to self-replicate to the highest degree possible

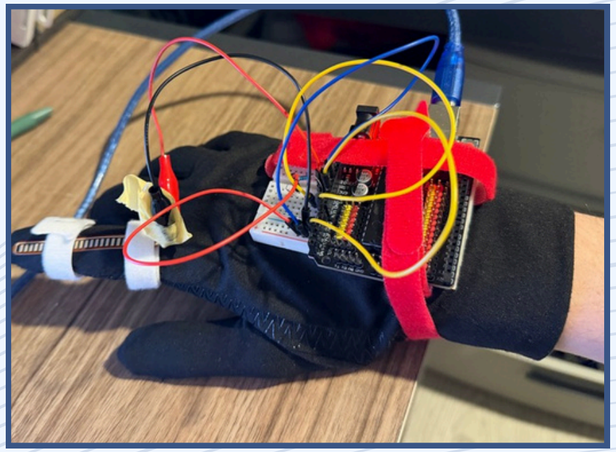
Overview

Robot Arm

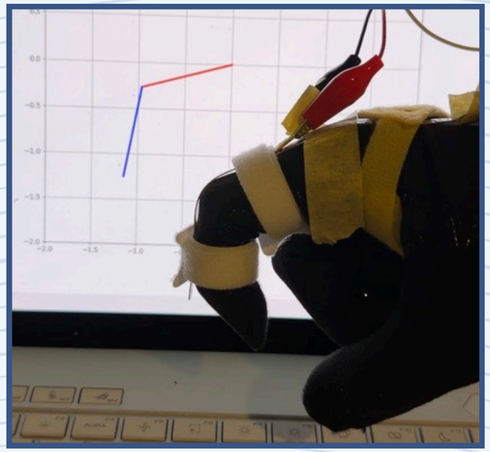
- Project to create a custom robot arm
- Hand-style end effector will allow a user to remotely manipulate objects
- Close collaboration between electical, mechanical, and software teams
- Personally designing hand-tracking glove
  - Significant signal processing challenge in addition to hardware design

Project Mitosis

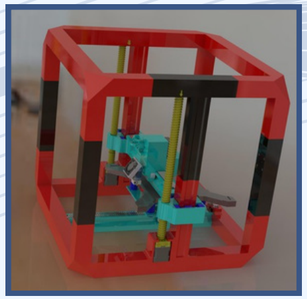
- FDM 3D printer designed with as many 3D printed components as possible
- Personal contributions include novel vertical axis movement system to improve precision
  - Design reduces issues caused by vertical resolution limits of 3D printed parts
  - Concept forms the core of sixth Mitosis printer design



One-joint prototype hand tracking glove using flex sensor

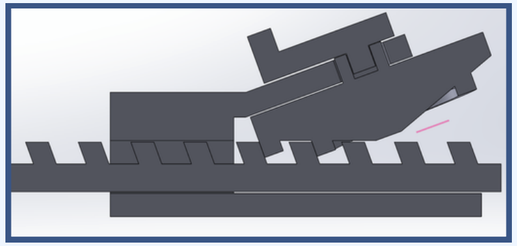


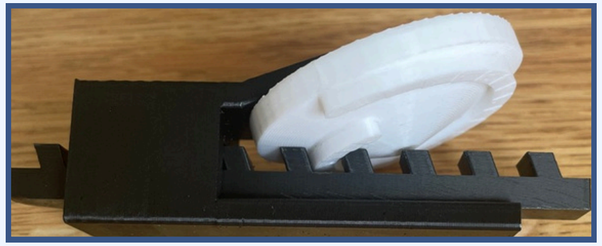
Two-joint finger tracking with live position plot



A render of a previous complete Mitosis printer design

Vertical Axis cross-section model and physical prototype



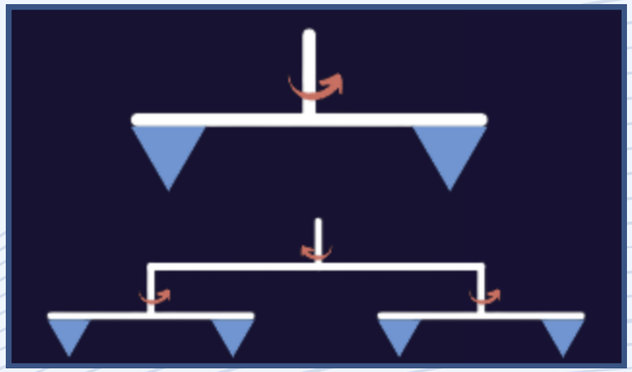


Parallelizing 3D Printing

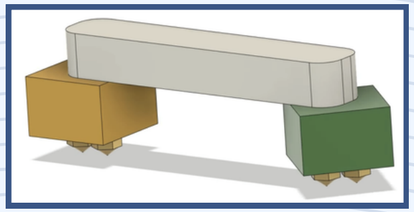
An independent research project using software simulation of multiple nozzles to improve the speed of 3D printers

Overview

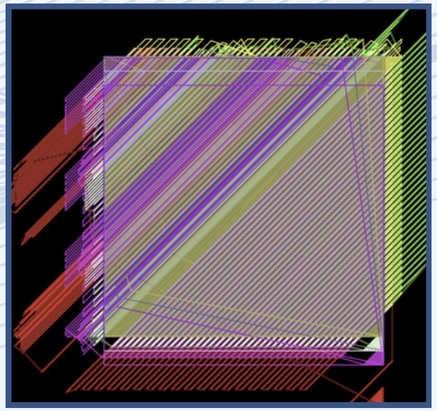
- FDM 3D Printing limited by speed of single printing nozzle
- Project enabled multiple nozzles to construct a single object simultaneously
- Efficient coordination of multiple nozzles was most significant challenge
- Two nozzles were set a fixed distance apart and assembly allowed to rotate
- Optimized path then found in software to prove concept viability



A diagram showing the rotational axes of the nozzles. Rotating each axis will change the distance between the nozzles relative to the direction of motion. This can be applied recursively, allowing for four or more independent nozzles



A 3D rendering of a notional 4-nozzle setup



The toolpath for one layer of a 4-nozzle print. Each color is the path of a different key point of the setup

Process

- Created slicing algorithm to efficiently use multiple nozzles while managing design constraints
- Program written in Java to reprocess existing GCODE into instructions for multi-nozzle printer
- System could be ‘nested’ for designs with 4 or more nozzles

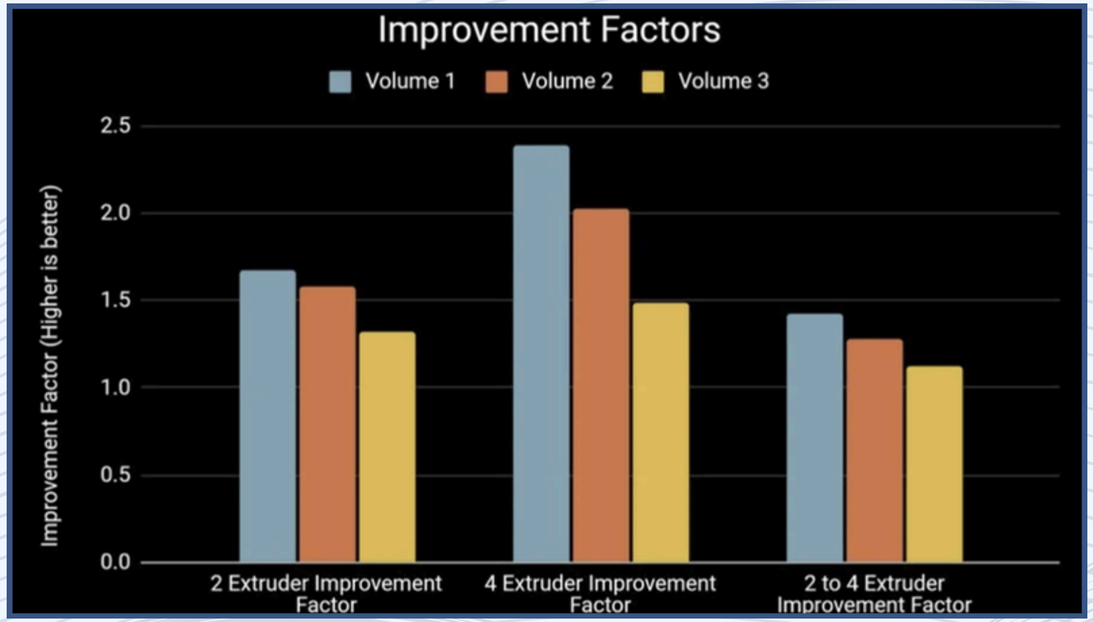
Parallelizing 3D Printing

Details

- Computer Science key to solving challenges
  - Memory management for simultaneous computer processes analogous to preventing interference between multiple nozzles
- Rotating assembly changes apparent distance between nozzles, allowing for simultaneous printing

Results

- Print speed improvement up to 2.4x
- Algorithm developed paths up to 80% of theoretical maximum performance
- Improvements continue to scale with more nozzles
- This project demonstrated viability of using multiple nozzles in the proposed configuration to greatly improve print speed
- Project won numerous awards, including second place at the 2023 Massachusetts Science and Engineering Fair.



Results of testing on three different models (Volume 1, 2, and 3) across different nozzle configurations. Each model was a rectangular prism with a different surface area to volume ratio, as the algorithm only applied to the interior volume (infill) of the model

Fan Automation at Ludlow Tennis Club

A system to cool the indoor courts without the expense and environmental impact of air conditioning

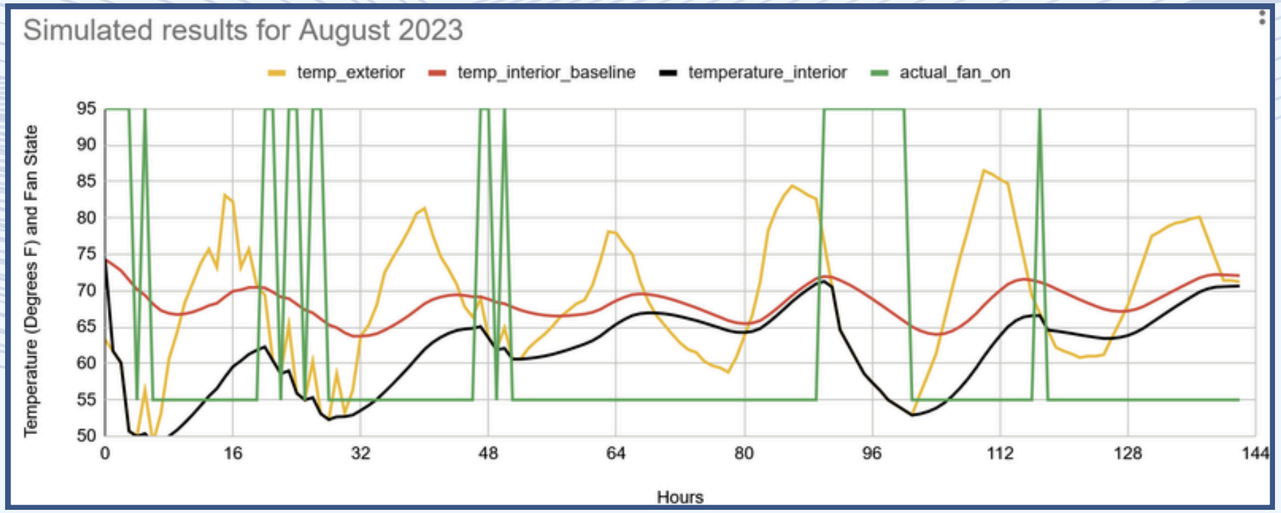
Overview

- New insulation caused condensation to form overnight during hot summer months
- Air conditioners and dehumidifiers use significant energy, presenting cost and environmental concerns
- I proposed that careful automated management of the ventilation fans could improve indoor conditions
- System required resilience in case of power failure, internet failure, or rain
- On testing data average temperatures improved by up to 9° F

The BME280 sensor used for determining indoor conditions

Details

- Integrated *Raspberry Pi* with temperature and humidity sensors using Python and accessed weather forecasts via a Web API
- Gathered data on heat transfer rates from the building under various outdoor conditions
- Accurately predicted interior conditions based on weather forecasts, enabling determination of optimal ventilation timing
- Dynamic system optimized scheduling up to one week in advance



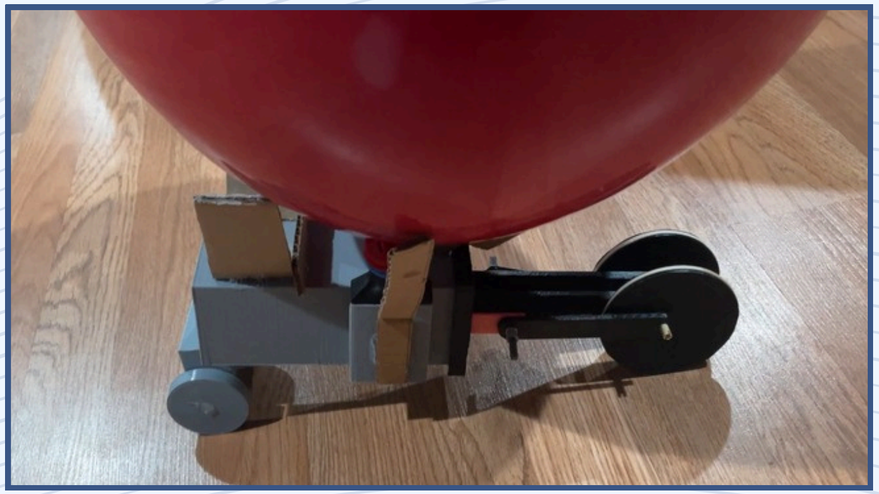
Simulated results from six days of data in August 2023. The difference between the red and black lines is the temperature improvement.
The green line is the fan state

3D Printed Compressed-Air Engine

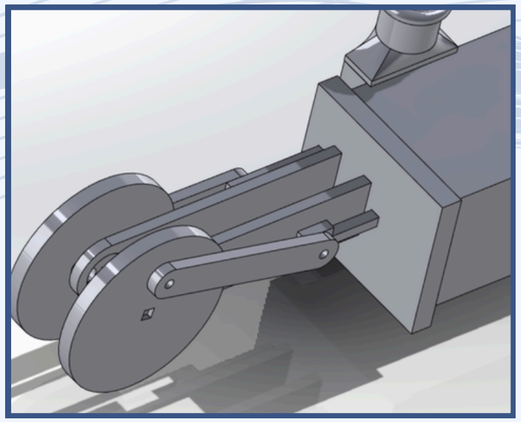
A class project to create a self-propelled 3D printed car

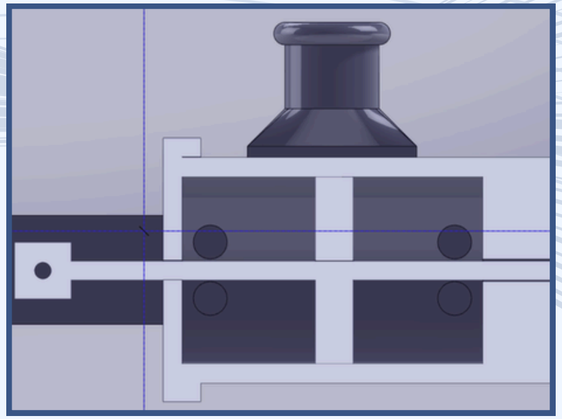
Overview

- Tasked with designing a 3D printed model car capable of being powered by a balloon
- Used a piston engine system
  - Design powered piston on both forward and reverse stroke
  - Major challenges include low air pressure in balloon and reducing friction while maintaining a seal in the piston and valve
- Designed and simulated system in SOLIDWORKS and printed using FDM 3D printers
- Final device achieved sustained propulsion on a level surface



The vehicle being prepared for a test





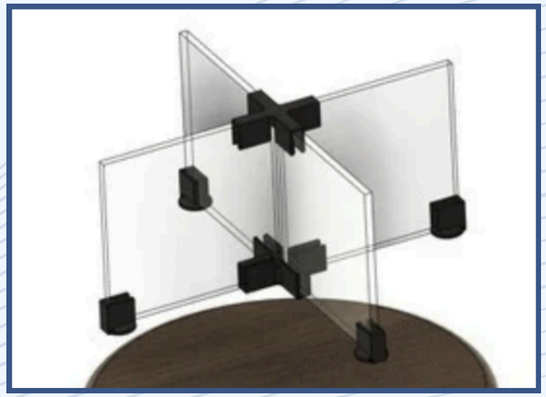
A rendering of the design in SOLIDWORKS, including a cutaway view of the piston

COVID-19 Protective Equipment

In the early days of the pandemic, protective equipment was in short supply. I designed and 3D printed 400+ custom parts for the installation of safety barriers in a local school.

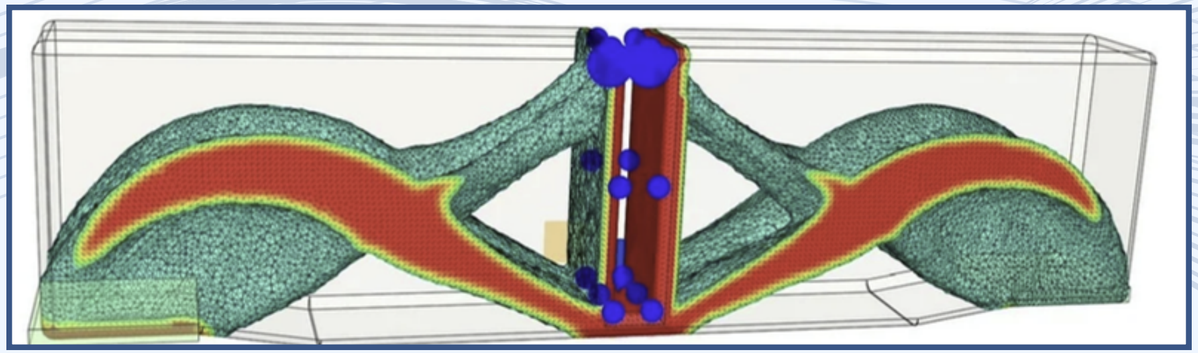
Overview

- As part of their reopening needs, Wilbraham & Monson Academy needed to install protective barriers on surfaces in a variety of spaces, including labs, classrooms, and the cafeteria.
- I designed and manufactured models including compliant fasteners and clips for dividers and shields
- Used CAD techniques including finite element analysis and topology optimization to increase quality and throughput
- Worked with clients and end users to rapidly incorporate feedback and iterate on designs



Rendering of 3D printed clips needed for a lunch table at Wilbraham & Monson

3D printed stands holding a divider at a desk



Results of the FEA study which optimized the geometry of the stand to reduce material usage, enabling increased production

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Availability
July-December 2026